Lecture 11 2025-10-07

$f(x) = \tan^{-1}(x)$ $f(0)=0$

$p_{n+1} = -g(p_n)$ $g(x) = (1+x^2)\tan^{-1}x - x$

$\exists$ nonzero fixed point p s.t. $g(p)=p$

$$2p = (1+p^2)\tan^{-1}p$$

If $p_0 = p$ then $p_1 = -g(p_0) = -p_0 = -p$

$$p_2 = -g(p_1) = -g(-p) = g(p) = p$$

$\Rightarrow$ $p_n = (-1)^n p$

$|p_0| > p$ $|g(p_0)| > |p_0|$

$\Rightarrow |p_1| > |p_0|$

$\vdots$

$|p_n| > |p_{n-1}|$

$p_1 = -g(p_0)$ $\rightarrow$ p_1 has opposite sign of p_0

$p_2 = -g(p_1)$ $\rightarrow$ p_2 has opposite sign of p_1

$q_n = (-1)^n p_n$ $\Rightarrow$ q_n all have same signs & increase monotonically

Similarly if $|p_0| < p$, the p_i decrease in absolute value but switch signs

$p_0 > 0$ $p_{2i+1} < 0$ $i \in \mathbb{N}$

$p_{2i} > 0$

$$q_{2i+1} = (-1)^{2i+1} p_{2i+1} = (-1) \cdot p_{2i+1} > 0$$

$$q_{2i} = (-1)^{2i} p_{2i} = p_{2i} > 0$$

2.4 Order of convergence (for iterative methods)

Suppose $p_n \to p$ & $p_n \neq p$ for $n \geq N$

Convergence of order α if

$$\lim_{n \to \infty} \frac{|p_{n+1} - p|}{|p_n - p|^\alpha} = \lambda > 0$$

Linear convergence: $\alpha = 1$, $\lambda < 1$

Quadratic convergence: $\alpha = 2$

Intuition: ① Suppose $\dfrac{|p_{n+1} - p|}{|p_n - p|} = \lambda < 1$ $\quad n = 1, 2, 3, \ldots$

Then $|p_n - p| = \lambda |p_{n-1} - p| = \lambda^2 |p_{n-2} - p| = \ldots = \lambda^n |p_0 - p|$

$= e^{n \log \lambda} |p_0 - p|$

② Suppose $\dfrac{|p_{n+1} - p|}{|p_n - p|^2} = \lambda$

$\Rightarrow$ we get super exponential convergence

$$|p_n - p| = \lambda |p_{n-1} - p|^2 = \lambda (\lambda |p_{n-2} - p|^2)^2$$

$$= \lambda^3 |p_{n-2} - p|^4 = \ldots$$

$$= \lambda^{2^n - 1} |p_0 - p|^{2^n}$$

So if $\beta = \lambda |p_0 - p| < 1$, then the result decays like β^{2^n}

$$\beta^{2^n} = e^{\log_\beta 2^n}$$

The terminology is confusing since a sequence that decays with rate $O(\frac{1}{n^r})$ e.g. $p_n = n^{-3}$ has $\alpha = 1, \lambda = 1$

$$\frac{|p_{n+1}|}{|p_n|} \overset{(p=0)}{=} \frac{\frac{1}{(n+1)^3}}{\frac{1}{n^3}} = \frac{n^3}{(n+1)^3} \longrightarrow 1$$

is not even linearly convergent in this terminology.

Theorem 2.8/2.9. Let $g \in C^1([a,b])$, $g(x) \in [a,b]$ $\forall x \in [a,b]$ $|g'(x)| \leq k < 1$ on $[a,b]$. Let p be the unique fixed point in $[a,b]$.

Then $p_{n+1} = g(p_n)$ either terminates exactly (after finite n) or

1) If $g'(p) \neq 0$, $p_n \to p$ only linearly ($\alpha = 1$, $\lambda \neq 0$)
2) If $g'(p) = 0$, then $p_n \to p$ quadratically (assume $g \in C^2([a,b])$)

Proof: If the iteration does not terminate exactly, then

Case 1: $p_{n+1} - p = g'(\zeta_n)(p_n - p)$

ζ_n between p_n and p_{n+1} (Taylor's theorem)

$$\lim_{n\to\infty} \frac{|p_{n+1}-p|}{|p_n-p|} = \lim_{n\to\infty} |g'(\zeta_n)| = |g'(p)| = \lambda \neq 0$$

($\zeta_n \to p$ & g' continuous)

$$\Rightarrow \quad \alpha > 1: \quad \lim_{n\to\infty} \frac{|p_{n+1}-p|}{|p_n-p|^{\alpha}} = \lim_{n\to\infty} \frac{|g'(\zeta_n)|}{|p_n-p|^{\alpha-1}} = \infty$$

forgot this earlier!

$\alpha - 1 > 0$
$|g'(\zeta_n)| \to |g'(p)|$
$|p_n - p| \to 0$

Case 2: $g(x) = \underbrace{g(p)}_{\| \; p} + \underbrace{g'(p)}_{\| \; 0}(x-p) + \frac{1}{2}g''(\xi)(x-p)^2$ (Taylor)

ξ is between x and p

$$g(x) = p + \frac{1}{2}g''(\xi_x)(x-p)^2$$

$$p_{n+1} = g(p_n) = p + \frac{g''(\xi_n)}{2}(p_n-p)^2 \quad \text{(pick } x = p_n \text{ above)}$$

ξ_n is between p and p_n

$$\lim_{n\to\infty} \frac{|p_{n+1}-p|}{|p_n-p|^2} = \lim_{n\to\infty} \frac{1}{2}|g''(\xi_n)| = \frac{1}{2}|g''(p)| = \lambda \text{ finite}$$

$g \in C^2([a,b])$

If $|p-p_0| < \delta$ sufficiently small then $\lambda|p-p_0| = \beta < 1$
and the error $|p-p_n| = O(\beta^{2^n})$ ∎

Remark: Recall that we can write Newton's method as a fixed point iteration with $g(x) = x - \frac{f(x)}{f'(x)}$ under the assumption that $f'(p) \neq 0$

Then $g'(x) = \frac{f(x)f''(x)}{(f'(x))^2}$

$f(p) = 0$, if $f'(p) \neq 0$, $f \in C^2$ then $g'(p) = 0$

$\Rightarrow$ Newton's method converges quadratically if $f'(p) \neq 0$

$$g'(x) = 1 - \frac{f'(x)}{f'(x)} + \frac{f(x)f''(x)}{(f'(x))^2} = \frac{f(x)f''(x)}{(f'(x))^2}$$

$$\frac{|p_{n+1}-p|}{|p_n-p|} = \lambda = 1 \Rightarrow |p_{n+1}-p| = |p_n-p| = \dots = |p_0-p|$$

Multiple roots

We needed $f'(p) \neq 0$ for Newton's method to work well:

Example: $f(x) = x^2$ $\quad f(0) = 0$ $\quad \Rightarrow p = 0$ with $f'(p) = 0$

$f'(x) = 2x$ $\quad f'(0) = 0$

Newton's iteration: $p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)} = p_n - \frac{p_n^2}{2p_n} = \frac{1}{2} p_n$

$p_{n+1} = \frac{1}{2} p_n = \frac{1}{4} p_{n-1} = \dots = \frac{1}{2^{n+1}} p_0$ $\Rightarrow$ linear convergence

$\left(\lim_{n \to \infty} \frac{|p_{n+1}|}{|p_n|} = \frac{1}{2} \right)$

Issue: $g(x) = x - \frac{f(x)}{f'(x)}$ $\qquad g'(x) = \frac{f''(x) f(x)}{(f'(x))^2} = \frac{2x^2}{4x^2} = \frac{1}{2} \neq 0$

does not satisfy condition for quadratic convergence in previous theorem

Definition 2.10 A solution p of $f(x) = 0$ is called a zero of multiplicity m of f if for $x \neq p$ we can write

$$f(x) = (x-p)^m q(x) \quad \text{where} \quad \lim_{x \to p} q(x) \neq 0$$

and q is continuous.

Theorem 2.12 Suppose $f \in C^m([a,b])$. The function f has a zero of multiplicity m at $x = p \in [a,b]$ iff

$f^{(j)}(p) = 0$ for $0 \leq j \leq m-1$

and $f^{(m)}(p) \neq 0$.

Proof: "$\Leftarrow$" By Taylor, we have $f(x) = \frac{1}{m!} f^{(m)}(\xi(x)) (x-p)^m$

ξ between x and

7 between x and p

$$= q(x)\,(x-p)^m$$

$$\Rightarrow q(x) = \frac{f(x)}{(x-p)^m} \quad \text{is continuous except possibly at } x=p$$

Since $q(x) = \frac{1}{m!} f^{(m)}(\xi(x)) \longrightarrow \frac{1}{m!} f^{(m)}(p) \neq 0$ as $x \to p$

($\xi(x) \to p$ as $x \to p$)

so q extends continuously to $x=p$ & $q(x) \neq 0$.